



A centroid-based gene selection method for microarray data classification



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HIGHLIGHTS

- A gene selection method is proposed by using centroid-based discriminant criterion.
- Kernel-based expectation is used to estimate the class centroid.
- An efficient algorithm is proposed for the L1-regularized optimization problem.
- Theoretical analysis indicates that the global optimal solution can be reached.
- Our method has linear complexity with the number of genes and samples.

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ABSTRACT

For classification problems based on microarray data, the data typically contains a large number of irrelevant and redundant features. In this paper, a new gene selection method is proposed to choose the best subset of features for microarray data with the irrelevant and redundant features removed. We formulate the selection problem as a L1-regularized optimization problem, based on a newly defined linear discriminant analysis criterion. Instead of calculating the mean of the samples, a kernel-based approach is used to estimate the class centroid to define both the between-class separability and the within-class compactness for the criterion. Theoretical analysis indicates that the global optimal solution of the L1-regularized criterion can be reached with a general condition, on which an efficient algorithm is derived to the feature selection problem in a linear time complexity with respect to the number of features and the number of samples. The experimental results on ten publicly available microarray datasets demonstrate that the proposed method performs effectively and competitively compared with state-of-the-art methods.

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1. Introduction

With the development of microarray techniques, a huge number of large-scale gene expression profiles have appeared. These microarray datasets are applied to various areas of application, such as tumor detection, cancer classification and the prediction of some rare diseases (Kurgan et al., 2001). Since these data usually have a lot of noisy and redundant information (Lu et al., 2014), it is desired to reduce the dimensionality of these data. Dimension reduction is often achieved by feature selection, which is the only method that maintains a direct relationship between a feature and a gene (Saey et al., 2007). For this reason, this process is often called gene selection

in the context of microarray data. In this paper, we focus on feature selection for the purposes of classification.

One popular strategy for feature selection is to select features such that the classes can be separated as much as possible with regard to the data space spanning (e.g., linear discriminant analysis), such as Fisher Score (Richard et al., 2001), Laplacian Score (He et al., 2005) and Trace Ratio (Nie et al., 2008). However, these traditional methods use the mean of samples as the centroid of the class to define the class separability measure, which is sensitive to the noisy data. Therefore, the classification accuracy of these methods can be affected notably when the data contains significant noise. Typically, microarray data has a huge number of irrelevant genes, which not only result in lower classification performance, but also add extra difficulties in finding informative genes (Oh et al., 2004; Bolón-Canedo et al., 2014). The LASSO method was proposed by Tibshirani, which adds L1 regularized

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penalty to the objective function (Tibshirani, 1996). And the L1 regularization technique has been proved to be able to lead to a sparse solution and it is usually deemed as an effective way to remove redundant and irrelevant features (Li et al., 2015). Nevertheless, it is difficult to solve the regularized problem efficiently due to the non-differentiability of L1 regularization (Liu et al., 2013). Moreover, how to obtain an optimal solution is another challenge for L1-regularized methods.

Recently, some L1-regularized logistic regression feature selection methods (Shi et al., 2010; Yuan et al., 2012; Tan et al., 2013) have been proposed, which can solve the L1 regularized problem efficiently. Although these methods are able to obtain the optimal solution under some special conditions, it is still challenging for them to identify the most discriminative subset from millions of features (Tan et al., 2013), due to the fact that they do not investigate local structures of sample. Actually, some existing methods that integrate some structures of sample (e.g., the margin of samples) for feature selection show excellent performance, such as Local Linear Feature Selection (LLFS) (Sun et al., 2010), which combines L1-regularized logistic regression with the large margin framework. However, LLFS has high computational cost when applied to multiclass problems. Microarray data classification poses a serious challenge for computational techniques, because of their large dimensionality (up to several tens of thousands of genes) with small sample sizes (Bolón-Canedo et al., 2014). To this end, it is necessary to consider the balance between the computational cost and classification accuracy for microarray data classification.

In this paper, we propose an efficient method to the feature selection problems for microarray data classification. The main contributions of this paper are as follows:

- We define an objective function for the optimization problem based on L1-regularized logistic regression and linear discriminant analysis (LDA) for the classes in microarray data. Instead of calculating the mean of the samples as the centroid of the class, we utilize the kernel-based expectation to estimate the centroid of the class, which is then used to define the between-class scatter and the within-class scatter. To obtain the global optimum efficiently, the L1 regularized logistic regression is incorporated into the objective function.
- We propose an efficient algorithm to solve the optimization problem by using Fletcher–Reeves conjugate gradient descent. Theoretical analyses of the algorithm suggest that the global optimal solution can be obtained with a general condition. An analysis of the method's computational complexity is conducted to demonstrate the efficiency of the method.
- Experiments on ten microarray datasets are conducted to evaluate the proposed method. The experimental results show the efficiency of the proposed method compared with the six state-of-the-art feature selection methods.

The rest of the paper is organized as follows: Section 2 reviews some related works about feature selection. Section 3 gives detailed description of the proposed method. Experimental evaluation is presented and discussed in Section 4. In Section 5, we conclude the paper and discuss the future work.

2. Related work

This section gives a brief review of existing methods that are relevant to our work. More details can be referred to (Yuan et al., 2012; Chandrashekar and Sahin, 2014).

According to the different selection strategies, feature selection methods can be divided into three types: filters, wrappers, and

embedded methods (Guyon and Elisseeff, 2003). Filter methods select features by information of the features and are independent of classifiers. For example, Laplacian Score (He et al., 2005), Trace Ratio (Nie et al., 2008) and Similarity Preserving Feature Selection (SPFS) (Zhao et al., 2013) use the class separability as a criterion to select the feature subsets. *F*-test (Le Cao et al., 2009) is one of the most popular filter methods, which is based on statistical hypothesis testing. These methods are simple and efficient. One weakness of these methods is their capability to deal with noises and outliers. Wrapper methods usually use the error probability of a classifier as the evaluation function to select features, such as Genetic Algorithms (GA) (Ooi and Tan, 2003) and successive feature selection (SFS) (Sharma et al., 2012). Wrapper algorithms typically perform better than filter methods while they have high computational complexity due to the fact that they need to train a large number of classifiers. In embedded methods, feature selection process is incorporated into the machine learning algorithm of a classifier. Popular embedded methods include Recursive Feature Elimination (RFE) (Guyon et al., 2002) and Extensions of SVM-RFE (MSVM-RFE) (Zhou and Tuck, 2007). Computational complexity is a major issue for embedded methods when the number of features is huge (Sun et al., 2010).

To deal with irrelevant and redundant features, one common strategy is to use L1 regularization (Guo and Guo, 2015). Popular L1-regularized methods include LASSO (Tibshirani, 1996) and L1-regularized logistic regression (Ng, 2004), are built based on the linearity assumption of data models. Clearly, such an assumption limits their applicability to general problems (Sun et al., 2010). Furthermore, L1 regularized methods for high-dimensional data has long been considered a difficult computational problem since the objective function is non-differentiable (Liu et al., 2013). In Wang et al. (2007), a Hybrid Huberized SVM (HHSVM) was proposed by compositing L1 and L2 regularization to form a more structured regularization. But it was designed only for binary problems. In Argyriou et al. (2007), this type of regularization was designed for the multiclass feature selection. In Nie et al. (2010), Robust Feature Selection (RFS) was proposed for the L2, 1 regularization and some experiments showed its efficiency and fast convergence rate. However, RFS does not investigate local structures of sample. In Liu et al. (2013), Large Margin Subspace Learning (LMSL) was proposed by combining subspace learning with large margin. Yet, LMSL can only obtain suboptimal solution.

Fisher linear discriminant analysis (LDA) (Fisher, 1936) is a classical technique for supervised dimensionality reduction, which seeks a projection transformation such that the between-class scatter is maximized and meanwhile the within-class scatter is minimized. Traditional methods considered various selection criteria and optimization techniques, such as Kullback–Leibler divergence (Fu et al., 1970) and Bhattacharyya distance (Chen, 1975). One main weakness of these methods is that they are able to handle only large-sample cases, where the number of observations is much larger than that of features. In the past decades, a large number of extensions to LDA (Ye et al., 2004; Li et al., 2006; Sugiyama, 2007) were developed to address the problems of small sample size (SSS) and/or rank limit. LDA can also be applied for feature selection under the general graph-based feature selection framework, such as Laplacian Score, Trace Ratio and SPFS. In this framework, the feature subset is selected based on the score of the entire feature subset, and the score is calculated in a trace ratio form Nie et al. (2008). Most of these LDA methods are based on L2-norm, which is likely to make the effect of outliers magnified (Wang et al., 2014).

Recently, several L1-regularized LDA methods (Li et al., 2015; Zhong and Zhang, 2013; Wang et al., 2014) have been developed in classification problems for high-dimensional data and have shown

promising results. These methods can solve the L1 regularized optimization problem efficiently by using some iterative techniques. However, most of these methods can only obtain local optimum instead of global optimum.

Some recently developed L1-regularized logistic regression methods, such as Minimax Sparse Logistic Regression (MSLR) (Tan et al., 2013), L1-LR (Yuan et al., 2012) and LLFS, are able to achieve the global optimum under some special conditions. Nevertheless, these methods either do not investigate the local structure of sample or have high computational cost.

3. The proposed method

In this section, we present a new method to address the above-mentioned issues. This method uses the class separability as a criterion for feature selection and it is suitable for multiclass microarray data. The way we use the class separability, however, is different than those of the LDA, which are used for extracting features through feature vector transformation. In contrast, the proposed method is a general feature selection method, where each feature is scored according to the criterion and the features with high scores are selected. In addition, the formulations and optimization techniques of our method are different from the methods under the general graph-based framework, which calculate the score in a trace ratio form.

3.1. Class separability measure

Class separability measure is an independent criterion that evaluates the goodness of a feature or feature subset by exploiting the intrinsic characteristics of the training data without involving any mining algorithm (Liu and Yu, 2005). Popular class separability measures include the Bhattacharyya distance, divergence and scatter-matrix-based measures (Wang, 2008). Due to its simplicity, the scatter-matrix-based measure is often favored, which includes the within-class scatter matrix S_w , between-class scatter matrix S_b , and Total scatter matrix S_t . The scatter matrices are defined as:

$$\begin{cases} S_w = \frac{1}{n} \sum_{j=1}^g \sum_{i=1}^{n_j} (x_i^{(j)} - u_j)(x_i^{(j)} - u_j)^T, \\ S_b = \frac{1}{n} \sum_{j=1}^g n_j (u_j - u)(u_j - u)^T, \\ S_t = \frac{1}{n} \sum_{i=1}^n (x_i - u)(x_i - u)^T = S_w + S_b, \end{cases} \quad (1)$$

where $x_i^{(j)}$ denotes the i -th sample in the class j , g is the number of classes, u_j and u are the centroid of the class j and the whole data set, respectively, i.e., $u_j = \frac{1}{n_j} \sum_{i=1}^{n_j} x_i^{(j)}$ and $u = \frac{1}{n} \sum_{i=1}^n x_i$.

Large class separability means small within-class scattering but large between-class scattering. The commonly used measures include $\text{tr}(S_b)/\text{tr}(S_w)$ and $|S_b|/|S_w|$, where $\text{tr}(A)$ and $|A|$ denote the trace and determinant of a square matrix A , respectively (Wang, 2008). As mentioned above, traditional LDA-related methods used the mean of samples as the centroid of the class to define the between-class distance and the within-class distance, which is sensitive to noisy data. Moreover, as these methods implicitly assume a Gaussian distribution (the scattering of data is evaluated via the mean and variance) for each class, they cannot correctly evaluate the class separability when the data presents a non-Gaussian structure (Wang, 2008). Therefore, we develop a probabilistic model, in which each sample is treated as a potential centroid of the class, with the probability estimated by kernel

density estimation. Instead of calculating the mean of the samples, the centroid of the class is calculated as an expectation of the samples.

Let $S = [x_1, \dots, x_n]^T \in \mathbb{R}^{n \times d}$ be a training data set containing n samples, where x_i is the i -th data sample with d features, $Y = [y_1, \dots, y_n]^T$ is the corresponding class labels, and $d \gg n$. The centroid of the class is defined as follows:

Definition 1. Let S be a training data set, x_i be the i -th sample from S . And $P(x_i = c_j)$ is the probability of x_i being the centroid of the class j . Let $u_j = \frac{1}{|H_j|} \sum_{x_i \in H_j} x_i$, where $H_j = \{x_i \mid 1 \leq i \leq n, y_i = j\}$ and $|H_j|$ are the number of samples in the class j . Then, the centroid of the class j is:

$$c_j = \sum_{x_i \in H_j} P(x_i = c_j) x_i = \sum_{x_i \in H_j} \frac{k(\|x_i - u_j\|_2)}{\sum_{x_i \in H_j} k(\|x_i - u_j\|_2)} x_i, \quad (2)$$

where $k(\cdot)$ is a kernel function. As the kernel function $k(d) = \exp(-d/\sigma)$ has been used in many methods (e.g., LMSL, LLFS), we also use the kernel function in this paper, where the kernel width σ is an input parameter. Note that if $\sigma \rightarrow +\infty$, we have $v_j = u_j$, which means that the mean of the samples is a special case of the centroid. Other kernel functions may also be used, and the details of their properties can be found in Atkeson et al. (1997).

In order to calculate the weight (score) of each feature, we define $w = [w_1, w_2, \dots, w_d] \in \mathbb{R}^{1 \times d}$ be a weight vector. And based on the definition of the centroid, the between-class distance and the within-class distance are defined as:

$$S_b = d_w(c_i, c_j) = \sum_{k=1}^d w_k |c_{ik} - c_{jk}| \quad (3)$$

and

$$S_w = \sum_{j=1}^g \sum_{x_i \in H_j} d_w(x_i, c_j) = \sum_{j=1}^g \sum_{x_i \in H_j} \sum_{k=1}^d w_k |x_{ik} - c_{jk}|, \quad (4)$$

where g is the number of classes, x_{ij} , c_{ij} the j -th element of x_i and c_i , respectively and $d_w(\cdot, \cdot)$ is a distance function about w . Varieties of distances $d_w(\cdot, \cdot)$ have been proposed in some references (Liu et al., 2013; Sun et al., 2010). For the purpose of this paper, $d_w(\cdot, \cdot)$ is defined as the following:

$$d_w(x, y) = \|x - y\|_w, \quad (5)$$

with $\|U\|_w = \sum_i w_i |U_i|$, $1 \leq i \leq d$, $w_i \geq 0$.

3.2. Objective function

For clarity, we first consider binary problems, while in Section 3.4 we generalize our algorithm to solve multiclass problems. In the same spirit as LDA, which maximizes the between-class distance while minimizing the within-class distance, our objective function is given by:

$$\min_w J(w) = S_w - \gamma S_b, \quad (6)$$

where γ is the balance coefficient to make the second term of the function can get the same range of the first term. So, γ can be set equal to n (n is the number of all samples).

The objective function (6) can be simplified as:

$$\min_w J(w) = - \sum_{i=1}^n w z_i^T, \quad (7)$$

where $z_i = (|x_{i1} - c_{j1}| - |c_{11} - c_{21}|, \dots, |x_{id} - c_{jd}| - |c_{1d} - c_{2d}|)$, with x_{ij} , c_{ij} the j -th element of x_i and c_i , respectively, and $x_i \in H_j$.

In order to transform (7) into a convex optimization problem, we introduce logistic regression model to formulate the new objective function as:

$$\min_w K(w) = \sum_{i=1}^n \log(1 + \exp(-wz_i^T)). \quad (8)$$

In applications with a huge number of features, generally, it is expected that most features are irrelevant. For example, in cancer prognosis, only several genes are involved in tumor spread and growth (Van't Veer et al., 2002; Wang et al., 2005). In order to obtain the sparse solution, one common strategy is to add an L1 regularization of w to the objective function. With the L1 regularization, the optimization problem is given by:

$$\min_w L(w) = \sum_{i=1}^n \log(1 + \exp(-wz_i^T)) + \lambda \|w\|_1, \quad (9)$$

where λ is the regularization parameter to control the sparseness of the solution. Note that one main difference between our method and other L1-regularized logistic regression methods (e.g., MSLR, LLFS) is that we use a novel class separability measure to define z_i of (9). Therefore, our method is able to identify a more discriminative subset from the features. Although LLFS use the large margin framework to define z_i , it has high computational cost due to the fact that it needs to compute margins of samples in each iteration.

3.3. An efficient algorithm to solve the objective function

To solve the optimization problem, we reformulate (9) slightly as:

$$\begin{cases} \min_w L(w) = \sum_{i=1}^n \log(1 + \exp(-wz_i^T)) + \lambda \|w\|_1 \\ s. t. w \geq 0. \end{cases} \quad (10)$$

Two reasons for the constraint $w \geq 0$ are:

- The constraint has been previously used in many methods (e.g., LLFS).
- The introduction of the constraint enables us to obtain an optimal solution.

Direct Gradient Method (DGM) (Cai et al., 2010) is an efficient algorithm for L1-regularized learning problems. The basic idea of DGM is to transform a convex optimization problem with a non-differentiable objective function into an unconstrained non-convex problem, and then solve the problem by gradient descent techniques. Inspired by DGM, we rewritten (10) as:

$$\min_v F(v) = \sum_{i=1}^n \log\left(1 + \exp\left(-\sum_j v_j^2 z_{ij}\right)\right) + \lambda \|v\|_2^2, \quad (11)$$

where $w_j = v_j^2$, $1 \leq j \leq d$ and z_{ij} is the j -th element of z_i . Then, the solution of v can be found through gradient descent. In this paper, the Fletcher–Reeves conjugate gradient descent (FR) method (Fletcher, 1997) was used to enhance the performance of the algorithm. The update rule is given by:

$$v^{(k+1)} = v^{(k)} + \alpha^{(k)} d^{(k)}, \quad k=0, 1, \dots, \quad (12)$$

where k is the number of iterations, and the step length $\alpha^{(k)}$ is obtained by carrying out some line search. The direction $d^{(k)}$ is defined by:

$$d^{(k)} = \begin{cases} -g^{(k)}, & k=0, \\ -g^{(k)} + \beta d^{(k-1)}, & k>0, \end{cases} \quad (13)$$

where $\beta = \frac{\|g^{(k)}\|}{\|g^{(k-1)}\|}$, and $g^{(k)}$ is the gradient of $F(v)$ at $v^{(k)}$.

To guarantee the convergence of the algorithm, we use the exact line search. The **global convergence** of FR method with exact line search has been proved in Powell (1984). Other line search methods (e.g., the Strong Wolfe line search (Dai and Yuan, 2000)) with FR method can also guarantee the global convergence.

In our algorithm, if some elements of u are extremely small (e.g., less than 10^{-5}), the corresponding features would be eliminated due to that it has a negligible impact on the subsequent iterations and the final solution found (Cai et al., 2010).

The following theorem shows that a global minimizer will be obtained with a **non-zero initial point** in our algorithm.

Theorem. Let $F(v)$ be a function of v defined in (11). If $\frac{\partial F}{\partial v}|_{v=v^*} = 0$, then v^* is not a local minimize, but a global minimize or a saddle point of $F(v)$. Moreover, if v^* is found through gradient descent with an initial point $v_i^{(0)} \neq 0$, $1 \leq i \leq d$, then v^* is a global minimizer of $F(v)$.

Proof. According to Sun et al. (2010), there is:

If $f(x)$ is a strictly convex function of $x \in R^d$ and $h(x) = f(y)$, where $y = [y_1, \dots, y_d] = [x_1^2, \dots, x_d^2]$. If $\frac{\partial h}{\partial x}|_{x=x^*} = 0$ and x^* is found through gradient descent with an initial point $x_i^{(0)} \neq 0$, $1 \leq i \leq d$, then x^* is a global minimizer of $h(x)$.

In the light of Sun et al. (2010), for fixed z_i , the objective function $L(w)$ defined in (10) is a strictly convex function of w . And $F(v) = L(w)$, where $w_i = v_i^2$, $1 \leq i \leq d$. The proof is then complete.

The pseudo code of our algorithm is presented in Algorithm 1.

Algorithm 1. Regularized Logistic Regression for binary-problems

Initialization:

S is the training samples, Y is the sample labels, λ is the regularization parameter, σ is the kernel width and is the stopping criterion.

Set $v^{(0)} = w^{(0)} = [1, 1, \dots, 1]$, $k=0$ and $\theta=0.01$.

Output: Feature weights w

1. Compute c_1, c_2 using Eq. (2);
2. Compute $F^{(0)} = F(v^{(0)})$ using Eq. (11);
3. **Repeat**
4. Compute $d^{(k)}$ using Eq. (13);
5. Update $v^{(k+1)} = v^{(k)} + \alpha^{(k)} d^{(k)}$, where $\alpha^{(k)}$ is determined via the exact line search;
6. **if** $v_i^{(k+1)} < 10^{-5}$, then
7. $v_i^{(k+1)} = 0$;
8. **end if**
9. $k = k + 1$;
10. **Until** $\|F^{(k)} - F^{(k-1)}\| < \theta$;
11. $w_i^{(k)} = (v_i^{(k)})^2$, $1 \leq i \leq d$.

3.4. Feature selection for multiclass problems

This section presents the extension of Algorithm 1 to multiclass problems. Firstly, we decompose the g -class problem into g binary sub-problems by using OVA techniques. Next, we calculate the feature weights by Algorithm 1 for each binary sub-problem. Finally, the feature ranking is implemented on the averages of the g feature weights. The pseudo code is presented in Algorithm 2.

Algorithm 2. Regularized logistic regression for multiclass problems

Initialization: S is the training samples, Y is the sample labels, λ is the regularization parameter, σ is the kernel width and θ is the stopping criterion. Set $W = [0, 0, \dots, 0]$, $g = \text{number of}$

categories and $\theta=0.01$.

Output: Feature weights W

1. **For** $r=1$ to g do
2. Update Y so that the r -th class with positive labels and all other classes with negative labels
3. Calculate feature weights W_r by calling $W_r = \text{Algorithm1}(S, Y, \lambda, \sigma, \theta)$
4. $W = W + W_r$
5. **End**

3.5. Computational complexity

There are two main parts in Algorithm 1, including to compute class separability based on the centroid and to solve the L1 optimization problem. For these two parts, the computational complexities are $O(nd)$ and $O(Ind)$ respectively. Here, n is number of samples, d is the feature dimensionality and I is the number of iterations. For comparison, the computational complexity of LLFS for binary problems is $O(I(n^2d + nd))$.

For Algorithm 2, it runs g iterations and for each iteration, the computational complexity is the same as that of Algorithm 1. Here, g is the number of classes and $g \geq 3$. Therefore, the computational complexity of Algorithm 2 is $O((nd+Ind)g)$.

4. Experiments and results analysis

In this section, several experiments are conducted to demonstrate the effectiveness of the proposed method for multiclass feature selection. All experiments are implemented with MATLAB (R2010a edition) and run in PC (Microsoft Windows 7, Intel Core (TM) i5-3317U, CPU 1.70 GHz, 4 GB of RAM).

4.1. Experimental data

The proposed Regularized Logistic Regression (RLR) method is applied on the ten publicly available microarray datasets, and all of them have the following characteristics: (1) the datasets are typical high-dimensional data. The dimensions of the five datasets are over 10,000 and two of them have more than 10 categories. (2) Research on some of these datasets has been very active. For example, the NCI60 and GCM datasets are recognized as difficult datasets by many researchers in the literature. (3) The number of features is much larger than the number of samples. (4) These datasets contain a huge number of redundant and irrelevant features. More details of the data sources can be found in Arizona State University feature selection repository (available at <http://www.featureselection.asu.edu>), the Kent Ridge Bio-medical Dataset (available at <http://datam.i2r.a-star.edu.sg/datasets/krbd/index.html>) and Cancer Program Datasets (available at <http://www.broadinstitute.org/cgi-bin/cancer/datasets.cgi>). In addition, we normalize the datasets by transforming the values of the datasets to zero mean and unit standard deviation. Table 1 summarizes the information of these datasets.

4.2. Experimental setup

The experiments use the classification accuracy as the main criterion to compare different methods. In order to avoid “selection bias”, we use 5-fold Cross Validation (CV) in our experiments on all microarray datasets. The samples of each dataset are randomly portioned into 5 equal size subsamples, 4 subsamples of which are

Table 1

Summary of datasets used in the experiments.

Dataset	#Instances	#Features	#Classes
CLL-SUB-111	111	11340	3
Breast	95	4869	3
Lung	203	12600	5
Tumors-11	174	12533	11
Stjude	327	12558	7
SRBCT	83	2308	4
NCI60	61	5244	8
GCM	198	16063	14
Colon	62	2000	2
DLBCL	77	7129	2

held out for training and the remaining one are used for testing. The process was repeated five times, with each of the 5 subsamples used once as the testing data. All comparison methods are implemented on the same dataset that is randomly generated by 5-fold CV. The process of 5-fold CV is repeated 20 times to obtain the averaged performance. Unlike some classifier-based method (i.e., single variable classifier method), which may be highly sensitive to the choice of classifiers (Fakhraei et al., 2014), the proposed method is independent of classifier (filter method). In the evaluation process, we only focus on the impact from feature selection methods rather than different classifiers. Therefore, we only use SVM (Linear kernel function, default parameters) as a baseline classifier to estimate the classification accuracy by using the features selected by each method.

4.3. Compared methods and parameter selection

The six state-of-the-art feature selection methods are chosen for the comparison purpose as the following:

- (1) RFS (Robust Feature Selection) (Nie et al., 2010).
- (2) LLFS (Local Linear Feature Selection) (Sun et al., 2010).
- (3) MSVM-RFE (Multiclass SVM-RFE) (Zhou, and Tuck, 2007).
- (4) F -test (Le Cao et al., 2009).
- (5) KernelPLS (Kernel Partial Least Squares) (Sun et al., 2014)
- (6) mRMR (minimum Redundancy Maximum Relevance) (Peng et al., 2005)

RFS is an efficient feature selection method with emphasizing joint L2, 1 regularized minimization on both loss function and regularization. LLFS is closely related with the proposed RLR, as RLR and LLFS both use L1 regularized logistic regression to formulate the objective function. As SVM-based methods have been widely used in various experiments, we implemented the OVA-based MSVM-RFE proposed in Zhou, and Tuck (2007). We found that by using the setting of Duan et al. (2007), MSVM-RFE has better performance. Since with high-dimensional datasets and the MSVM-RFE method is time consuming, we adopt the strategy used in Guyon et al. (2002), which removes the features ranked the lowest 50% from each iteration recursively. F -test is one of most popular filter methods and is based on statistical hypothesis testing. KernelPLS is a very efficient method based on partial least squares (PLS) and theory of Reproducing Kernel Hilbert Space. The number of components of KernelPLS is determined by 5-fold CV. mRMR is of filter model, which removes redundant features via considering pairwise feature correlation measured by mutual information. The regularization parameter λ of the three methods (RFS, LLFS, and RLR) is determined by 5-fold CV. The kernel width σ of RLR and LLFS is also tuned with 5-fold CV.

4.4. Analysis of the results based on 5-fold CV

Our goal is to select a minimum number of genes to achieve a highly accurate prediction performance. In our experiments, we take the number of features from 1 to maximum number of 50 features, which is also used in Sun et al. (2010).

4.4.1. Average accuracy of classification

The averaged accuracy results in Table 2 are obtained by SVM using top 1, 2, 3,...,50 features selected for the five different methods (RLR, RFS, MSVM-RFE, LLFS, and *F*-test).

It can be observed in Table 2 that the proposed RLR has an advantageous average accuracy. Among the seven methods, the proposed RLR yields the best results in six out of the ten datasets, and has no significant differences (p -value ≥ 0.05) compared with the best ones when tested on other three datasets. The classification accuracy averaged on all datasets is shown on last row. RLR has a 0.4% higher classification accuracy than LLFS, which also incorporates L1 regularized logistic regression into the objective function. Compared with other methods, RLR performs better with 0.8–8.4% higher accuracy.

4.4.2. Classification accuracy of different features

To further investigate the classification performance of different methods under the different number of selected features, we plotted the experimental results in Fig. 1. As seen in Fig. 1, for the Lung and NCI60 datasets, the proposed RLR has the better classification accuracy. The average accuracy of RLR is usually higher than other methods. For CLL-SUB-111, Tumors-11 and Colon datasets, RLR can obtain a higher accuracy when the number of selected features is small, which demonstrates that RLR has stronger ability to remove redundant features. In addition, the proposed RLR usually has a smaller standard deviation on most datasets which indicates that our method has better robustness for microarray datasets or depends less on training set.

4.4.3. Evaluation of computational efficiency

The proposed RLR is computationally efficient. Table 3 shows the CPU time of seven methods, averaged over the 20 runs. As can be seen from the table, in terms of computational efficiency, our method performs the second best on four datasets (CLL-SUB-111, Breast, Lung and DLBCL) and performs the third best on other five datasets. RLR is slower than RFS on some datasets because it needs to compute the within-class distance and the between-class distance in order to obtain better performance for feature selection.

MSVM-RFE, LLFS and *F*-test are more time consuming than RLR on most datasets.

4.4.4. Sensitivity of the parameters

Fig. 2 shows that the insensitivity of RLR with respect to the regularization parameter λ . It can be observed that the average accuracy results of RLR in most datasets do not obviously change with different values of λ chosen from {0.1, 0.5, 1, 5, 10}.

Fig. 3 shows that the insensitivity of RLR with respect to the kernel width σ . It can be observed that the average accuracy results of RLR do not obviously change on most datasets with different values of σ chosen from {50, 100, 200, 300, 600}.

4.5. Statistical significance test

For a rigorous comparison, a paired two-tailed *t*-test is performed. The *p*-value of the *t*-test represents the probability that two sets of compared results come from distributions with an equal mean. *P*-value ≤ 0.05 is considered statistically significant. We use the previous 20 random results of 5-fold CV to verify the statistical difference in terms of classification accuracy for different methods. The comparison is implemented on the same training set and the testing set generated by 5-fold CV.

The results are shown in Table 4. As shown in Table 4, the classification accuracy of the proposed RLR is significantly better than those of RFS in six datasets. In three datasets (SRBCT, Colon, DLBCL), the two methods have no significant difference. For RLR and MSVM-RFE, RLR is significantly better than MSVM-RFE in four datasets (Lung, CLL-SUB-111, SRBCT and GCM) while they have no significant differences in other datasets. For RLR and LLFS, RLR is significantly better than LLFS in four datasets. And the two methods have no significant differences in the rest datasets except Breast. RLR is significantly better than *F*-test in all datasets except Colon. For RLR and KernelPLS, RLR is significantly better than KernelPLS in seven datasets while they have no significant differences in the rest datasets. RLR is significantly better than mRMR in all datasets except Tumors-11.

4.6. Discussion

In this section, we provide further discussions on the proposed RLR method. As illustrated by the classification results, RLR performs better than RFS on six datasets, indicating the importance of investigating local structures of sample. Although LLFS uses the margin framework, it does not perform as well as RLR on three datasets, suggesting the limitation of using the margin framework in feature selection. Furthermore, LLFS is even 10 times slower

Table 2
Average classification accuracy and standard deviations (%) of SVM.

Dataset	RLR	RFS	MSVM-RFE	LLFS	<i>F</i> -test	KernelPLS	mRMR
CLL-SUB-111	74.67(5.67)	73.55(6.09)	73.26(6.02)	74.63(5.67)	57.65(6.64)	73.64(6.92)	71.01(5.98)
Breast	60.39(7.34)	63.11(8.55)	59.07(7.97)	62.80(8.38)	56.97(8.97)	58.89(7.29)	55.32(7.89)
Lung	93.26(2.85)	91.91(2.88)	91.63(2.71)	91.92(3.05)	85.38(2.27)	91.27(2.63)	92.27(2.44)
Tumors-11	81.86(4.03)	78.83(3.60)	81.24(3.42)	80.46(3.64)	57.61(4.35)	68.93(4.57)	81.22(3.61)
Stjude	85.38(2.44)	83.27(2.71)	85.25(2.15)	85.00(2.49)	75.39(4.38)	76.67(2.40)	84.60(2.58)
SRBCT	97.94(1.52)	97.71(1.90)	97.58(1.64)	98.01(1.46)	95.53(1.85)	96.42(2.24)	95.83(2.52)
NCI60	59.21(9.83)	56.40(9.69)	57.57(10.48)	58.81(10.30)	55.12(10.41)	55.14(9.72)	54.66(10.08)
GCM	61.61(5.22)	60.72(4.60)	52.42(5.01)	60.29(4.91)	51.46(6.64)	49.84(4.98)	56.19(4.78)
Colon	84.69(8.56)	85.63(8.03)	84.06(7.60)	83.48(7.50)	83.95(6.92)	84.83(7.72)	81.29(8.03)
DLBCL	93.04(4.76)	92.97(5.00)	93.47(5.07)	92.95(5.20)	88.91(6.48)	88.51(6.80)	91.66(5.62)
AVG	79.21	78.41	77.56	78.84	70.80	74.41	76.41

The boldfaced values are the highest ones or the ones without significant difference to the highest (p -value ≥ 0.05). The value in the parentheses is standard deviations (%).

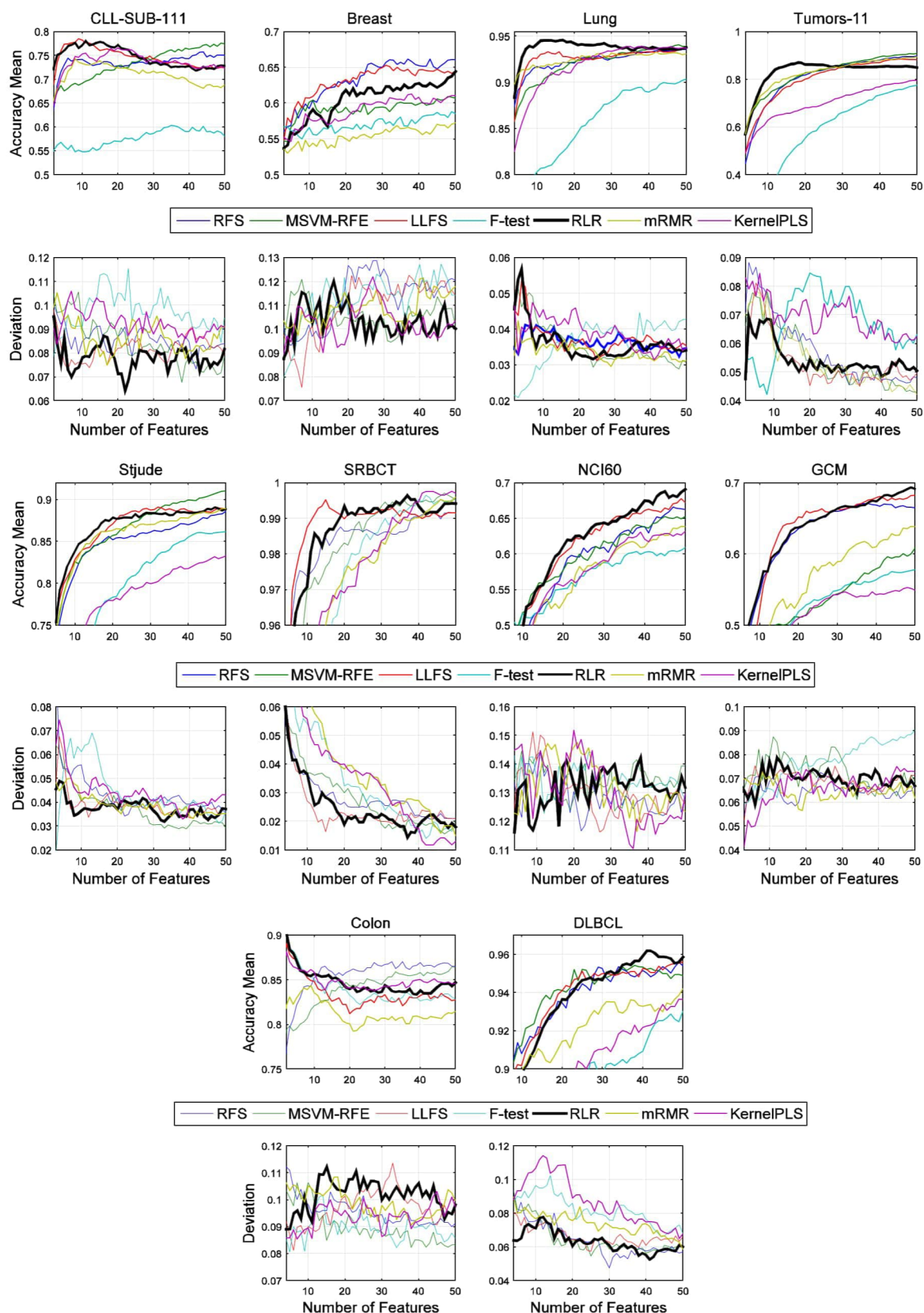


Fig. 1. Relationship between feature dimension and recognition rate (average accuracy and standard deviation) (20 random results of 5-fold CV).

than RLR on most datasets. The MSVM-RFE method, which uses RFE to deal with noise and redundant features, does not perform as well as RLR on four datasets. Moreover, it should be noticed that

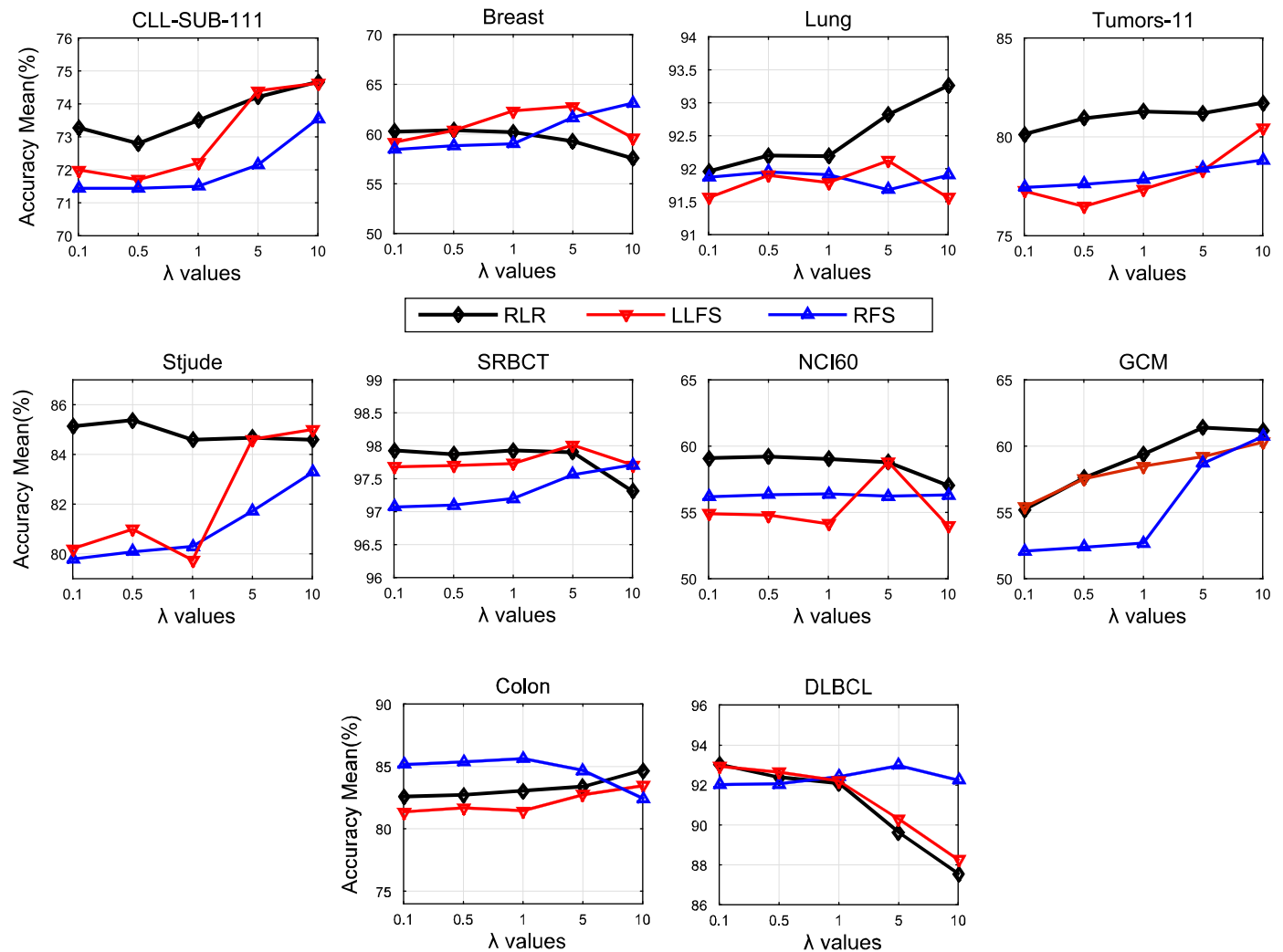
RLR requires much less computational time than MSVM-RFE on most datasets. One main issue of *F*-test is its capability of addressing the noise and outliers. RLR significantly outperforms

Table 3

CPU time per run (in seconds) of five algorithms performed on eight datasets.

Dataset	RLR	RFS	MSVM-RFE	LLFS	F-test	KernelPLS	mRMR
CLL-SUB-111	10.56	11.52	84.50	168.70	51.01	2.38	36.18
Breast	4.03	4.11	30.25	22.05	42.85	0.98	22.78
Lung	31.18	32.11	118.96	482.53	63.25	5.17	51.11
Tumors-11	45.07	25.87	253.12	1355.02	73.68	5.28	54.31
Stjude	60.64	81.45	680.90	2311.8	72.72	12.07	46.32
SRBCT	3.40	1.62	12.61	21.90	12.43	0.51	22.36
NCI60	7.96	2.59	39.62	58.77	26.24	0.63	36.28
GCM	83.18	33.35	489.01	858.24	104.35	9.28	44.97
Colon	1.11	1.03	5.43	3.55	9.04	0.28	23.85
DLBCL	2.72	4.06	15.71	37.49	34.16	0.87	22.79
Average	24.98	19.77	173.01	532.00	48.97	3.75	36.10

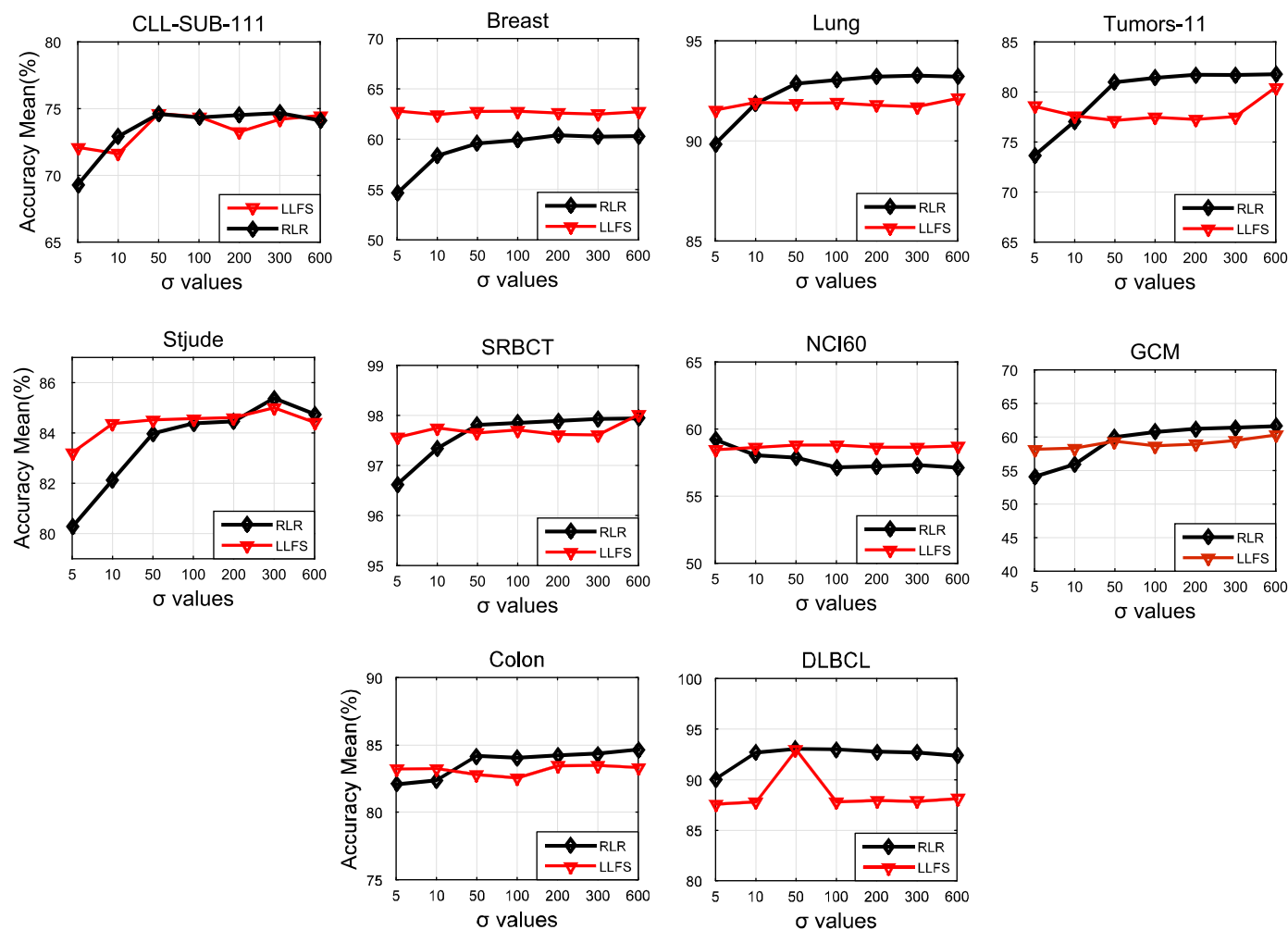
The boldfaced values are the best ones.

**Fig. 2.** Sensibility of the regularization parameter λ .

F-test on most datasets. AS KernelPLS lacks a mechanism to deal with noise and outliers, RLR significantly outperforms KernelPLS on seven datasets. The mRMR method removes redundant features via considering pairwise feature correlation measurement; however, mRMR does not investigate local structures of sample. RLR outperforms mRMR on all datasets except Tumors-11. In summary, compared to the six state-of-the-art methods, RLR is competitive and works effectively for microarray datasets.

5. Conclusion

In this paper, we first discussed the problems faced by the feature selection methods in classifying microarray data. This problem becomes difficult due to considerable redundant and irrelevant features in the data. Different from traditional LDA-related methods, we utilize the kernel-based expectation to estimate the centroid of the class, which is then used to define the class separability measure. Moreover,

Fig. 3. Sensitivity of the kernel width σ .Table 4
Statistical significance test.

Dataset	RLR vs. RFS	RLR vs. MSVM-RFE	RLR vs. F-test	RLR vs. LLFS	RLR vs. KernelPLS	RLR vs. mRMR
CLL-SUB-111	(+) 0.0335	(+) 0.0478	(+) $1.83e-36$	(=) 0.9285	(=) 0.1384	(+) $2.56e-9$
Breast	(-) $2.27e-4$	(=) 0.1298	(+) $1.64e-5$	(-) $2.90e-4$	(=) 0.0612	(+) $3.16e-9$
Lung	(+) $2.07e-8$	(+) $1.57e-10$	(+) $3.97e-37$	(+) $2.25e-9$	(+) $9.75e-15$	(+) $8.74e-7$
Tumors-11	(+) $5.15e-13$	(=) 0.1168	(+) $4.96e-75$	(+) $4.62e-6$	(+) $2.04e-47$	(=) 0.1666
Stjude	(+) $3.40e-12$	(=) 0.6032	(+) $4.82e-54$	(=) 0.0834	(+) $9.52e-49$	(+) 0.0031
SRBCT	(=) 0.1773	(+) 0.0370	(+) $6.50e-34$	(=) 0.4693	(+) $1.66e-14$	(+) $1.82e-16$
NCI60	(+) $6.46e-4$	(=) 0.0606	(+) $5.63e-6$	(=) 0.5836	(+) $1.70e-6$	(+) $3.71e-7$
GCM	(+) 0.0338	(+) $5.34e-27$	(+) $2.77e-25$	(+) $1.31e-4$	(+) $1.33e-43$	(+) $6.77e-20$
Colon	(=) 0.1082	(=) 0.3380	(=) 0.2211	(+) 0.0465	(=) 0.8013	(+) $6.22e-6$
DLBCL	(=) 0.8516	(=) 0.2936	(+) $1.44e-12$	(=) 0.8048	(+) $7.07e-13$	(+) 0.0015
Summary	6(+), 3(=), 1(-)	4(+), 6(=), 0(-)	9(+), 1(=), 0(-)	4(+), 5(=), 1(-)	7(+), 3(=), 0(-)	9(+), 1(=), 0(-)

In the table (+) implies that our method is statistically better than the compared method, (-) implies the contrary and (=) means that the two methods have no significant differences. The p-value associated is also given.

L1-regularized logistic regression is incorporated into the objective function and we proved that the global optimal solution can be reached under a general condition by using the gradient descent technique. Experimental results show that our method is effective and competitive compared with four state-of-the-art methods.

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